

Time series. ARIMA models

Modelització Estadística

Grau en Intel·ligència Artificial

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Outline

- 1 Stochastic Models
- 2 **AR(p)** models
- 3 **MA(q)** models
- 4 **ARMA(p,q)** models
- 5 **ARIMA(p,d,q)** models
- 6 **SARIMA(p, d, q)(P, D, Q)[s]** models

Time series. Stochastic Models

General Stochastic model for a time series:

$$X_t = G(X_{t-1}, X_{t-2}, \dots, Z_{t-1}, Z_{t-2}, \dots) + Z_t \quad Z_t \sim WN(0, \sigma_z)$$

The most easy mathematical function $G(\cdot)$ is the linear combination of the components:

Linear Stochastic model for a time series

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \dots + Z_t \quad Z_t \sim WN(0, \sigma_z)$$

$$\phi_1, \phi_2, \dots, \theta_1, \theta_2, \dots \in \mathbf{R}$$

The variable at time t is a **linear combination of past observations and disturbances plus a new disturbance independent from the past.**

AR(p) models

Autoregressive model (AR)

A **pth-order autoregressive model**, or AR(p), takes the form:

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \dots + \phi_p X_{t-p} + Z_t$$

$$(1 - \phi_1 B - \phi_2 B^2 - \dots + \phi_p B^p) X_t = Z_t$$

where:

X_t is stationary and $\phi_1, \dots, \phi_p$ are the parameters (constants)

Z_t is a Gaussian white noise ($E(Z_t) = 0$, $V(Z_t) = \sigma_Z^2$)

B is the Backshift operator: $BX_t = X_{t-1}$

p is the lag of the farthest observation included

An **AR(p) model** is a regression model with lagged values of the dependent variable in the independent variable positions, hence the name **Auto-Regressive model**.

AR(p) models

- If $\mu \neq 0$, then

$$(X_t - \mu) = \phi_1(X_{t-1} - \mu) + \cdots + \phi_p(X_{t-p} - \mu) + Z_t$$

or

$$X_t = \alpha + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + Z_t$$

where the constant term is:

$$\alpha = \mu(1 - \phi_1 - \cdots - \phi_p)$$

Or a more concise expression:

$$\phi_p(B)X_t = Z_t$$

where $\phi(B) = (1 - \phi_1 B - \cdots - \phi_p B^p)$

Characteristic autoregressive polynomial

$\phi_p(B)$ is the **characteristic autoregressive** polynomial of order p .

AR(p) models

AR(1) process, considering $\mu = 0$:

$$(1 - \phi B)X_t = Z_t, \quad Z_t \sim N(0, \sigma_Z^2).$$

Or equivalently:

$$X_t = \phi X_{t-1} + Z_t, \quad Z_t \sim N(0, \sigma_Z^2).$$

Autocovariance function derivation:

$$\gamma(0) = E[X_t^2] = E[(\phi X_{t-1} + Z_t)^2] = \phi^2 \gamma(0) + \sigma_Z^2 \Rightarrow \gamma(0) = \frac{\sigma_Z^2}{1 - \phi^2}$$

$$\gamma(1) = E[X_t X_{t-1}] = E[(\phi X_{t-1} + Z_t) X_{t-1}] = \phi \gamma(0)$$

⋮

$$\gamma(h) = E[X_t X_{t-h}] = E[(\phi X_{t-1} + Z_t) X_{t-h}] = \phi \gamma(h-1) = \phi^h \gamma(0)$$

Reminder: The noise at time t is independent of the past:

$$E[Z_t X_s] = E[Z_t Z_s] = 0 \quad s < t.$$

AR(p) models

AR(1) process: $(1 - \phi B)X_t = Z_t$

Autocorrelation function:

$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \phi$$

:

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = \phi^h$$

Recursion: $\rho(h) = \phi\rho(h-1)$

Partial Autocorrelation function:

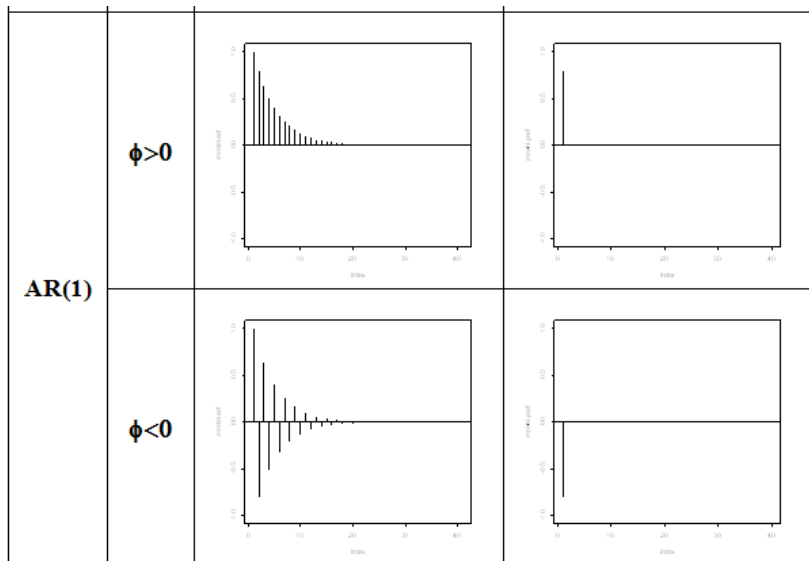
$$\phi_{1,1} = \phi$$

:

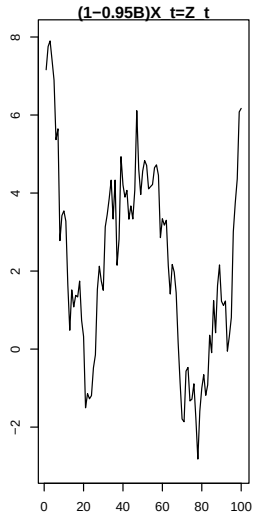
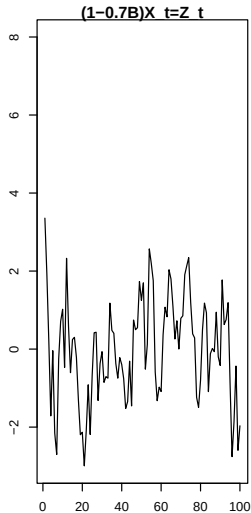
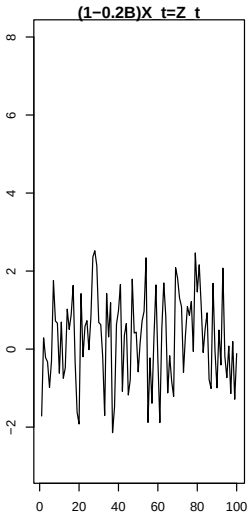
$$\phi_{h,h} = 0 \quad h > 1$$

AR(p) models

AR(1) process: $(1 - \phi B)X_t = Z_t$



AR(1) models



AR(p) models

AR(2) process: $(1 - \phi_1 B - \phi_2 B^2)X_t = Z_t$

Considering $\mu = 0$, then

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t \quad Z_t \sim N(0, \sigma_Z^2)$$

Autocovariance function:

$$\gamma(0) = E[X_t^2] = (\phi_1^2 + \phi_2^2)\gamma(0) + 2\phi_1\phi_2\gamma(1) + \sigma_Z^2$$

$$\gamma(1) = E[X_t X_{t-1}] = \phi_1\gamma(0) + \phi_2\gamma(1)$$

:

$$\gamma(h) = E[X_t X_{t-h}] = \phi_1\gamma(h-1) + \phi_2\gamma(h-2) \quad h > 1$$

MA(q) models

Moving average model (MA)

A **qth-order moving average model**, or MA(q), takes the form:

$$X_t = z_t + \theta_1 Z_{t-1} + \theta_2 Z_{t-2} + \cdots + \theta_q Z_{t-q}$$

In other words,

$$X_t = \theta(B)Z_t$$

where

$$\theta(B) = 1 + \theta_1 B + \theta_2 B^2 + \cdots + \theta_q B^q$$

is the moving average operator or characteristic polynomial.

An **MA(q) model** is a regression model with the dependent variable, X_t , depending on previous values of the errors rather than on the variable itself.

MA(q) models

MA(1) process, considering $\mu = 0$:

$$X_t = (1 + \theta B)Z_t, \quad Z_t \sim N(0, \sigma_Z^2)$$

Or equivalently:

$$X_t = Z_t + \theta Z_{t-1}, \quad Z_t \sim N(0, \sigma_Z^2)$$

Autocovariance function derivation:

$$\gamma(0) = E[X_t^2] = E[(Z_t + \theta Z_{t-1})^2] = (1 + \theta^2)\sigma_Z^2$$

$$\gamma(1) = E[X_t X_{t-1}] = E[(Z_t + \theta Z_{t-1})(Z_{t-1} + \theta Z_{t-2})] = \theta \sigma_Z^2$$

:

$$\gamma(h) = E[X_t X_{t-h}] = 0 \quad h > 1$$

MA(q) models

MA(1) process: $X_t = (1 + \theta B)Z_t$.

Autocorrelation function:

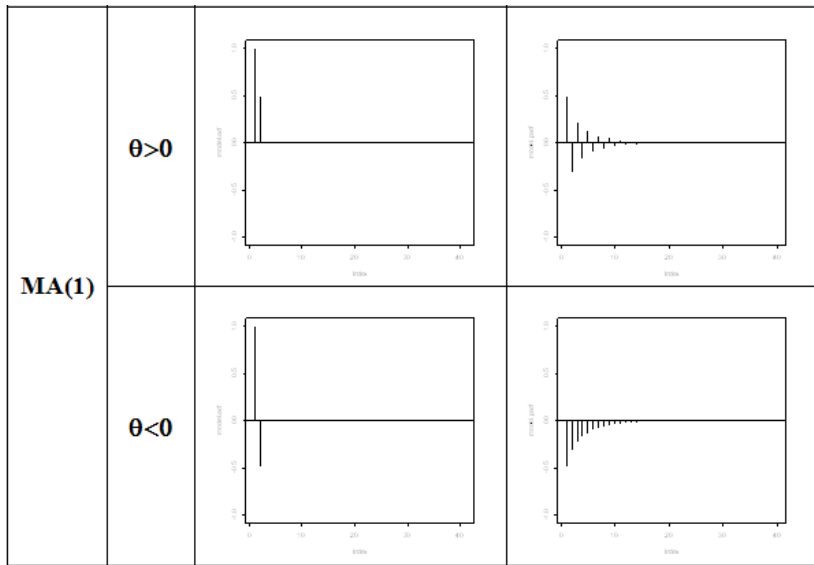
$$\rho(1) = \frac{\gamma(1)}{\gamma(0)} = \frac{\theta}{1 + \theta^2}$$

$$\vdots$$

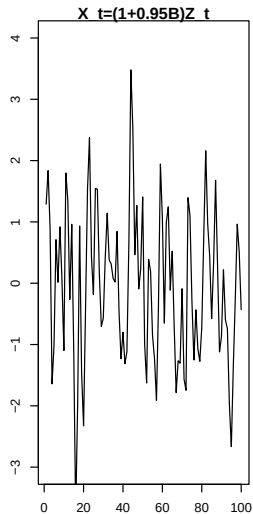
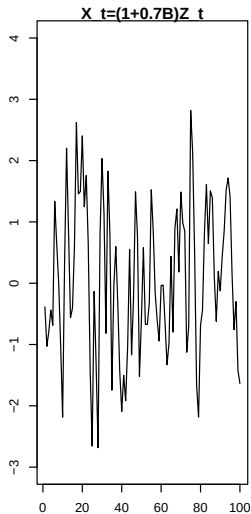
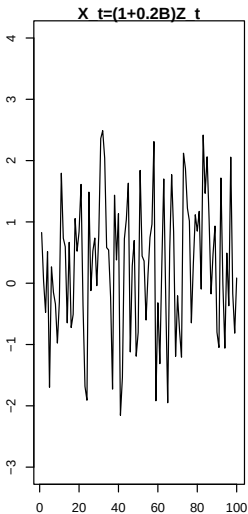
$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} = 0 \quad h > 1$$

MA(q) models

$$\text{MA}(1): x_t = (1 + \theta B)z_t$$



MA(1) models



ARMA(p,q) models

AutoRegressive Moving Average model (ARMA)

A times series $\{X_t; t = 0, \pm 1, \pm 2, \dots\}$ is an **AutoRegressive Moving Average model**, ARMA (p,q), if it is stationary and

$$X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

with $\phi_p \neq 0$, $\theta_q \neq 0$ and $\sigma_Z^2 > 0$

The parameters p and q are called the autoregressive and the moving average orders, respectively.

ARMA(p,q) models

If X_t has a nonzero mean μ and $\alpha = \mu(1 - \phi_1 - \dots - \phi_p)$, the model will be:

$$X_t = \alpha + \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

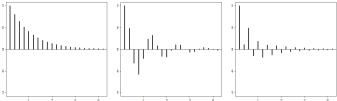
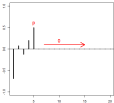
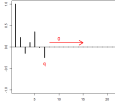
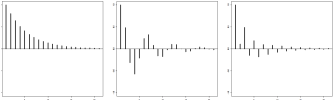
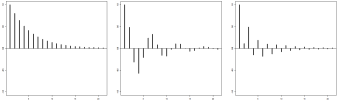
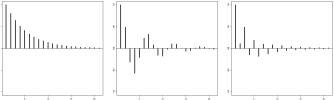
When $q = 0$, the model is an AR(p).

When $p = 0$, the model is a MA(q)

ARMA(p,q) model can be written in concise form as,

$$\phi(B)_p X_t = \theta_q(B) Z_t$$

Identification of ARMA(p,q) models

MODEL	ACF	PACF
AR(p)	Decreasing patterns (∞ lags not null) 	p = Last lag not null 
MA(q)	q = Last lag not null 	Decreasing patterns (∞ lags not null) 
ARMA(p,q)	Decreasing patterns (∞ lags not null) 	Decreasing patterns (∞ lags not null) 

ARMA(p,q) models

Expression of an $ARMA(p, q)$ model, using characteristic polynomials:

$$\phi(B)x_t = \theta(B)z_t$$

Under certain conditions, $ARMA(p, q)$ models can be expressed as an $AR(\infty)$ (**invertibility**) or $MA(\infty)$ model.

Expression as an $AR(\infty)$:

$$\frac{\phi(B)}{\theta(B)}x_t = \pi(B)x_t = z_t$$

Expression as an $MA(\infty)$:

$$x_t = \frac{\theta(B)}{\phi(B)}z_t = \psi(B)z_t$$

Stationarity and Invertibility of ARMA(p,q) models

Expression of an $ARMA(p, q)$ model, using characteristic polynomials:

$$(1 - \phi_1 B - \dots - \phi_p B^p)x_t = (1 + \theta_1 B + \dots + \theta_q B^q)z_t$$

Under certain conditions, $ARMA(p, q)$ models can be expressed as a pure $AR(\infty)$ (**invertibility**) or $MA(\infty)$ process.

Expression as an $AR(\infty)$:

$$\frac{1 - \phi_1 B - \dots - \phi_p B^p}{1 + \theta_1 B + \dots + \theta_q B^q} x_t = (1 - \pi_1 B - \pi_2 B^2 - \dots) x_t = z_t$$

Expression as an $MA(\infty)$:

$$x_t = \frac{1 + \theta_1 B + \dots + \theta_q B^q}{1 - \phi_1 B - \dots - \phi_p B^p} z_t = (1 + \psi_1 B + \psi_2 B^2 + \dots) z_t$$

Stationarity and Invertibility Conditions

For **AR**, **MA**, and **ARMA** models, stationarity and invertibility depend on the **roots of the characteristic polynomials**:

Stationarity

An **AR(p) model** is stationary if the **roots of the AR characteristic polynomial** lie **outside the unit circle**.

- AR characteristic polynomial: $\phi(B) = 1 - \phi_1 B - \dots - \phi_p B^p$
- Condition: $|\text{roots}| > 1$

MA(q) models are **always stationary**

Invertibility

A **MA(q) model** is invertible if the **roots of the MA characteristic polynomial** lie **outside the unit circle**.

- MA characteristic polynomial: $\theta(B) = 1 + \theta_1 B + \dots + \theta_q B^q$
- Condition: $|\text{roots}| > 1$

AR(p) models are **always invertible**

Stationarity and Invertibility Conditions

ARMA Models:

- An **ARMA(p, q)** model is **stationary** if the AR part's roots lie outside the unit circle and **invertible** if the MA part's roots lie outside the unit circle.

Ensuring stationarity and invertibility helps create stable, reliable models for time series forecasting.

ARIMA Models

ARIMA Models

An **ARIMA(p, d, q)** model generalizes ARMA models by introducing **differencing** to achieve stationarity:

- **Autoregressive (AR(p))**: Uses past values of the series to predict the current value.
- **Differencing (I(d))**: Makes the series stationary by differencing.
- **Moving Average (MA(q))**: Models the error term as a linear combination of past error terms.

The ARIMA model equation is given by:

$$\nabla^d X_t = \phi_1 X_{t-1} + \dots + \phi_p X_{t-p} + Z_t + \theta_1 Z_{t-1} + \dots + \theta_q Z_{t-q}$$

where $\nabla^d X_t$ represents the differenced series.

When to Use an ARIMA Model

Key Points:

- 1 **Non-stationary series:** ARIMA is suitable for time series that show trends or seasonal patterns, which require differencing.
- 2 **Parameter selection:**
 - p : Order of autoregressive part
 - d : Degree of differencing
 - q : Order of moving average part

Example

If a series requires **one difference** to become stationary and exhibits characteristics of an AR(1) and MA(1), an **ARIMA(1,1,1)** model would be appropriate.

The differencing term d is key in ARIMA, as it transforms non-stationary data into stationary, making it suitable for ARMA modeling.

Seasonal ARIMA (SARIMA) Models

SARIMA models

To handle **seasonal patterns**, we extend ARIMA to **SARIMA(p, d, q)(P, D, Q)[s]**:

- **(P, D, Q)**: Seasonal autoregressive, differencing, and moving average terms.
- **s**: Season length (e.g., 12 for monthly data with annual seasonality).

The SARIMA model equation is:

$$\nabla^d \nabla_s^D X_t = \phi(B) \Phi(B^s) X_t + \theta(B) \Theta(B^s) Z_t$$

where:

- ∇^d : Non-seasonal differencing.
- ∇_s^D : Seasonal differencing.
- $\phi(B)$ and $\Phi(B^s)$: Non-seasonal and seasonal AR terms.
- $\theta(B)$ and $\Theta(B^s)$: Non-seasonal and seasonal MA terms.

SARIMA models

Key Points:

- 1 **Seasonal Patterns:** SARIMA is ideal for data with recurring patterns, such as monthly sales or quarterly temperatures.
- 2 **Parameter selection:**
 - P, D, Q : Seasonal AR, differencing, and MA terms
 - s : Season length (defines the periodicity of the season)

Example

For a monthly time series with yearly seasonality, a **SARIMA(1,1,1)(1,1,1)[12]** model could be appropriate, indicating both regular and seasonal differencing and dependencies. SARIMA models capture both **short-term fluctuations** and **long-term seasonal cycles**.

R

- See script [02_ts_simula.R](#)